

Project Title: Automated Radiology Report Scribe

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Domain Research Note: Radiology Narrative Generation

1. Problem Statement & Objective Manual transcription of imaging findings bottlenecks radiology workflows, driving specialist burnout and diagnostic latency. The objective is to automate preliminary clinical narratives specifically the identification and description of visual anomalies using generative artificial intelligence.

2. Technical Architecture: Generative VLMs The solution requires a Vision Language Model (VLM) for multimodal understanding:

- Vision Encoding: A convolutional neural network (CNN) backbone (e.g., ResNet) extracts hierarchical, semantic visual features from medical scans.
- Text Generation: The model maps these visual features into a language embedding space to conditionally generate context-aware clinical text.

3. Dataset Infrastructure: MultiCaRe The open source MultiCaRe dataset serves as the foundational data pipeline.

- Utility: It provides synchronized text/image pairs (radiological images linked directly to expert written clinical reports).
- Application: This establishes an immediate, authoritative ground truth for training and validation, bypassing the need for manual data scraping.

4. Evaluation Framework (Ablation Study) The core deliverable is a comparative performance analysis measuring how effectively the model adapts to the clinical lexicon.

- Baseline: An off the shelf, pre trained VLM (e.g., LLaVA variant) evaluated zero shot.
- Fine-Tuned Model: The baseline VLM specifically fine-tuned on MultiCaRe text/image pairs.
- Success Metrics: Output variance will be quantified using standard Natural Language Generation (NLG) metrics (ROUGE, BLEU) to measure alignment against the ground-truth clinical descriptions.